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*      STAAD.Pro
*      Version 2007   Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=   JAN 18, 2013
*      Time=   16:29:30
*
*      USER ID:
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1. STAAD SPACE
INPUT FILE: 20121120 T-Goal Post 039 FR1.STD
2. START JOB INFORMATION
3. ENGINEER DATE 12-MAY-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 2 5 1 -5.73; 4 5 4.495 -5.73; 5 5 4.495 -4.98; 6 5 4.495 -6.65
9. MEMBER INCIDENCES
10. 3 2 4; 4 4 6; 5 4 5
11. DEFINE MATERIAL START
12. ISOTROPIC STEEL
13. E 2.05E+008
14. POISSON 0.3
15. DENSITY 76.8195
16. ALPHA 1.2E-005
17. DAMP 0.03
18. END DEFINE MATERIAL
19. MEMBER PROPERTY JAPANESE
20. 4 5 TABLE ST H428X407X20
21. 3 TABLE ST H428X407X20
22. CONSTANTS
23. BETA 90 MEMB 3
24. MATERIAL STEEL ALL
25. SUPPORTS
26. 2 FIXED
27. ***** DEAD LOAD *****
28. LOAD 1 LOADTYPE NONE TITLE DL
29. SELFWEIGHT Y -1
30. MEMBER LOAD
31. 4 CON GY -19.6 0.4
32. ***** FRICTIONAL LOAD *****
33. LOAD 2 LOADTYPE NONE TITLE FL
34. MEMBER LOAD
35. 4 CON GX -5.88 0.4
36. ***** WIND LOAD X-DIRECTION*****
37. LOAD 3 LOADTYPE NONE TITLE WLX
38. MEMBER LOAD
39. 4 5 UNI GX -0.364
40. 3 UNI GX -0.364
41. ***** WIND LOAD Z-DIRECTION*****
42. LOAD 4 LOADTYPE NONE TITLE WLZ
43. MEMBER LOAD
44. 3 UNI GZ -0.342
45. ***** NOTIONAL LOAD X-DIRECTION*****
46. LOAD 5 LOADTYPE NONE TITLE NLX
47. SELFWEIGHT X -0.015
48. MEMBER LOAD
49. 4 CON GX -0.294 0.4
50. *****NOTIONAL LOAD Z-DIRECTION*****
51. LOAD 6 LOADTYPE NONE TITLE NLZ
52. SELFWEIGHT Z -0.015
53. MEMBER LOAD

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54. 4 CON GX -0.294 0.4
55. ***** LOAD COMBINATION - SERVICE *****
56. LOAD COMB 7 1.0DL+1.0FL
57. 1 1.0 2 1.0
58. LOAD COMB 8 1.0DL+1.0FL+1.0WLX
59. 1 1.0 2 1.0 3 1.0
60. LOAD COMB 9 1.0DL+1.0FL+1.0WLX
61. 1 1.0 2 1.0 4 1.0
62. LOAD COMB 10 1.0DL+1.0FL+1.0NLX
63. 1 1.0 2 1.0 5 1.0
64. LOAD COMB 11 1.0DL+1.0FL+1.0NLZ
65. 1 1.0 2 1.0 6 1.0
66. ***** LOAD COMBINATION - ULTIMATE *****
67. LOAD COMB 12 1.0DL+1.4FL
68. 1 1.0 2 1.4
69. LOAD COMB 13 1.4DL+1.4FL
70. 1 1.4 2 1.4
71. LOAD COMB 14 1.0DL+1.4FL+1.4WLX
72. 1 1.0 2 1.4 3 1.4
73. LOAD COMB 15 1.0DL+1.4FL+1.4WLZ
74. 1 1.0 2 1.4 4 1.4
75. LOAD COMB 16 1.0DL+1.4FL+1.4NLX
76. 1 1.0 2 1.4 5 1.4
77. LOAD COMB 17 1.0DL+1.4FL+1.4NLZ
78. 1 1.0 2 1.4 6 1.4
79. PERFORM ANALYSIS

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PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 4/ 3/ 1

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 18

80. LOAD LIST 7 TO 11
81. PRINT JOINT DISPLACEMENTS ALL

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4	7	-0.1050	-0.0014	-0.0206	-0.0001	0.0083	0.0004
	8	-0.1244	-0.0014	-0.0206	-0.0001	0.0085	0.0005
	9	-0.1050	-0.0014	-0.0235	-0.0001	0.0083	0.0004
	10	-0.1124	-0.0014	-0.0206	-0.0001	0.0087	0.0005
	11	-0.1102	-0.0014	-0.0214	-0.0001	0.0087	0.0005
5	7	0.5162	0.0073	-0.0206	-0.0001	0.0083	0.0004
	8	0.5104	0.0073	-0.0206	-0.0001	0.0085	0.0005
	9	0.5162	0.0081	-0.0235	-0.0001	0.0083	0.0004
	10	0.5413	0.0073	-0.0206	-0.0001	0.0087	0.0005
	11	0.5420	0.0075	-0.0214	-0.0001	0.0087	0.0005
6	7	-0.8675	-0.0141	-0.0206	-0.0001	0.0083	0.0004
	8	-0.9037	-0.0141	-0.0206	-0.0001	0.0085	0.0005
	9	-0.8675	-0.0150	-0.0235	-0.0001	0.0083	0.0004

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11 -0.9109 -0.0144 -0.0214 -0.0001 0.0087 0.0005

***** END OF LATEST ANALYSIS RESULT *****

82. PRINT SUPPORT REACTION ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
2	7	5.88	33.91	0.00	8.23	-2.35	-20.55
	8	7.76	33.91	0.00	8.23	-2.40	-24.90
	9	5.88	33.91	1.20	10.32	-2.35	-20.55
	10	6.39	33.91	0.00	8.23	-2.48	-22.07
	11	6.17	33.91	0.21	8.73	-2.47	-21.58

***** END OF LATEST ANALYSIS RESULT *****

83. LOAD LIST 12 TO 17

84. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
3	12	2	33.91	0.00	8.23	-3.29	-28.77	8.23
		4	-24.23	0.00	-8.23	3.29	0.00	-8.23
	13	2	47.48	0.00	8.23	-3.29	-28.77	11.53
		4	-33.92	0.00	-8.23	3.29	0.00	-11.53
	14	2	33.91	0.00	10.86	-3.37	-34.86	8.23
		4	-24.23	0.00	-9.08	3.37	0.00	-8.23
	15	2	33.91	1.67	8.23	-3.29	-28.77	11.16
		4	-24.23	0.00	-8.23	3.29	0.00	-8.23
	16	2	33.91	0.00	8.94	-3.47	-30.90	8.23
		4	-24.23	0.00	-8.74	3.47	0.00	-8.23
	17	2	33.91	0.30	8.64	-3.46	-30.21	8.93
		4	-24.23	-0.10	-8.64	3.46	0.00	-8.23
4	12	4	0.00	22.15	8.23	0.00	-3.29	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
	13	4	0.00	31.01	8.23	0.00	-3.29	12.62
		6	0.00	0.00	0.00	0.00	0.00	0.00
	14	4	0.00	22.15	8.70	0.00	-3.51	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
	15	4	0.00	22.15	8.23	0.00	-3.29	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
	16	4	0.00	22.15	8.70	0.00	-3.48	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
	17	4	-0.05	22.15	8.64	0.00	-3.46	9.01
		6	0.00	0.00	0.00	0.00	0.00	0.00
5	12	4	0.00	2.08	0.00	0.00	0.00	0.78
		5	0.00	0.00	0.00	0.00	0.00	0.00
	13	4	0.00	2.91	0.00	0.00	0.00	1.09
		5	0.00	0.00	0.00	0.00	0.00	0.00

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5	0.00	0.00	0.00	0.00	0.00	0.00
15	4	0.00	2.08	0.00	0.00	0.78
	5	0.00	0.00	0.00	0.00	0.00
16	4	0.00	2.08	-0.04	0.00	0.78
	5	0.00	0.00	0.00	0.00	0.00
17	4	0.04	2.08	0.00	0.00	0.78
	5	0.00	0.00	0.00	0.00	0.00

***** END OF LATEST ANALYSIS RESULT *****

85. LOAD LIST ALL

86. PARAMETER 1

87. CODE BS5950

88. BEAM 2 ALL

89. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
3 ST	H428X407X20	PASS	ANNEX I.1	0.050	14
		33.91 C	34.86	8.23	-
4 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.023	13
		0.00	-3.29	12.62	0.00
5 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.002	13
		0.00	0.00	1.09	0.00

***** END OF TABULATED RESULT OF DESIGN *****

90. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= JAN 18,2013 TIME= 16:29:31 ****

* For questions on STAAD.Pro, please contact *

* Research Engineers Offices at the following locations *

* *

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* USA:	+1 (714)974-2500	support@bentley.com
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